

PRABHATH VINAY VIPPARTHI

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SUMMARY

Data Scientist and AI Engineer with 4 years of experience spanning enterprise data science and applied AI development. Built predictive models, forecasting systems, and customer analytics for enterprise clients, then expanded into NLP, embeddings, and retrieval-based AI systems through graduate study and applied engineering projects. Currently working at Scale AI preparing datasets, engineering features, evaluating classification and language models, and supporting the deployment and monitoring of production ML/AI pipelines. Comfortable moving between traditional statistical modeling and modern AI engineering, with working knowledge of Python, SQL, PySpark, MLflow, Docker, and cloud-based deployment workflows.

TECHNICAL SKILLS

- **Programming & Data Analysis:** Python (3.10/3.11), SQL, Pandas, NumPy, SciPy
- **Statistics & Machine Learning:** scikit-learn, PyTorch, TensorFlow, Logistic Regression, Random Forest, Gradient Boosting, Cross-Validation, ROC-AUC, Feature Engineering
- **NLP & Applied AI:** NLP, Transformers, Hugging Face, spaCy, Embeddings, LLMs
- **Generative AI & Retrieval Systems:** RAG, Vector Search, LLM-as-a-Judge, vLLM, Google GenAI (Gemini)
- **Big Data & Data Processing:** PySpark, Apache Spark, Databricks, DuckDB
- **Data Engineering & Pipelines:** Apache Airflow, dbt, ETL/Data Pipelines, Apache Arrow, JSONL
- **MLOps, Evaluation & Monitoring:** MLflow, Model Evaluation, Model Monitoring, Data Drift Detection
- **Databases & Vector Search:** PostgreSQL, pgvector, SQLite, SQLAlchemy, psycpg2
- **APIs & Application Development:** FastAPI, Uvicorn, REST APIs, tree-sitter
- **Cloud, Containers & Deployment:** Azure, AWS, Render, Docker, Docker Compose, Kubernetes
- **Visualization & Analytics:** Power BI, Tableau, Streamlit
- **Development Tools & CI/CD:** GitHub Actions, Google Colab, Amazon S3

PROFESSIONAL EXPERIENCE

Applied AI Engineer | Scale AI, USA

Jan 2026 – Present

- Reviewed business, data-quality, and AI requirements with stakeholders and built Python, SQL, and PySpark pipelines to process 10K+ records, cutting manual validation effort by 30%.
- Prepared training and evaluation datasets and engineered statistical and ML features using Pandas, NumPy, and scikit-learn to strengthen automated anomaly and data-quality detection.
- Built classification models in scikit-learn and PyTorch, applying cross-validation, precision/recall analysis, and threshold tuning to improve minority-class detection.
- Integrated Hugging Face transformers and embeddings for text classification and semantic evaluation, reducing selected manual review activity by 25%.
- Implemented embedding-based retrieval workflows and Python REST APIs to support similarity search and AI/LLM evaluation use cases with lower inference overhead.
- Supported model deployment using MLflow, Docker, Kubernetes, Azure, and Databricks, improving consistency across releases.
- Monitored model quality, data drift, latency, and pipeline failures, and optimized PySpark workloads to reduce recurring processing time by 28%.
- Created Power BI dashboards tracking AI/ML performance, data-quality trends, and business KPIs to support data-driven decisions.

Data Scientist | HCL Tech, India

Jun 2021 – Jul 2024

- Partnered with business stakeholders to translate retention and growth objectives into churn and forecasting use cases, scoping the analytical approach for each engagement.
- Conducted EDA and statistical analysis in Python, Pandas, NumPy, and SQL to uncover customer behavior patterns and key churn drivers.
- Engineered 30+ customer-level features covering purchase frequency, recency, transaction value, and engagement to strengthen predictive modeling inputs.

- Built and compared Logistic Regression, Random Forest, and Gradient Boosting classifiers using cross-validation and ROC-AUC, improving high-risk customer recall by 18%.
- Developed demand forecasting models from historical and seasonal patterns, reducing forecast error by approximately 15% and supporting operational planning.
- Processed 5K+ transaction records with PySpark and Databricks, cutting recurring analytical batch runtime by 35%.
- Automated recurring data and model workflows using Airflow, AWS, and MLflow, and delivered Tableau dashboards covering churn segments, forecasts, and business KPIs.

SELECTED AI, DATA SCIENCE & ENGINEERING PROJECTS

AI-Assisted Digital Badge Classification System — Graduate Capstone, NJIT LDI

- Built and deployed a badge classification system (FastAPI, React/Vite, SQLite) that replaced manual taxonomy tagging, processing three input formats through a multi-layer NLP signal-extraction pipeline combining lexicon patterns, regex rules, and a spaCy verb matcher.
- Designed a rule-based classification engine with confidence scoring, plain-English explanations, audit logging, and human-in-the-loop review; validated the system with a 351-test suite running in GitHub Actions CI.

FinSight — Pre-Market Intelligence Platform

- Built and deployed a market-briefing platform (FastAPI on Render) that ingests data for roughly 130 symbols each weekday, computes technical indicators, and generates AI-assisted summaries distributed by email through an automated GitHub Actions workflow.
- Implemented a retrieval-augmented Q&A layer over briefing history using text embeddings stored in PostgreSQL with pgvector, returning answers with source citations and similarity scores.
- Instrumented LLM calls with prompt, output, latency, and status logging for monitoring, and maintained 68 offline tests across 9 modules in CI.

StarCoder2 Self-Alignment Pipeline (Academic Project)

- Adapted a self-instruct methodology to TypeScript, streaming source files from a large public code dataset, extracting functions with tree-sitter, and filtering high-quality seed examples through automated validation checks.
- Ran a multi-stage generation pipeline on StarCoder2-3B via vLLM to produce instruction-response pairs, incorporating an LLM-as-a-judge stage to filter low-confidence outputs.

NYC Taxi Medallion Data Pipeline

- Built a medallion-architecture pipeline processing nearly 6 million NYC taxi trip records, applying PySpark data-quality rules and loading cleaned data into a DuckDB warehouse.
- Modeled dbt business marts with data-quality tests, orchestrated the workflow as a daily Airflow DAG, and validated it end-to-end in GitHub Actions CI.
- Built an interactive Streamlit dashboard to analyze revenue and payment trends across boroughs and time periods.

EDUCATION

Master of Science (M.S.) in Data Science, Computational Track

May 2026

New Jersey Institute of Technology (NJIT), Ying Wu College of Computing | Newark, NJ | GPA: 3.7/4.0

CERTIFICATIONS / PROFESSIONAL TRAINING

- AI Fluency for Builders — Anthropic Education
- Introduction to Agent Skills — Anthropic Education
- AI Fluency: Framework & Foundations — Anthropic Education
- Supervised Machine Learning: Regression & Classification — Coursera
- Google Cloud Network Engineer Professional — Google Cloud
- AWS Cloud Virtual Internship — AWS
- Salesforce Administrator Virtual Internship — Salesforce